

```
1 import java.io.*;
2 import java.util.*;
3
4 public class Zoo {
5     private String zooName;
6     private ArrayList<Animal> animals;
7
8     public Zoo(String zooName) {
9         this.zooName = zooName;
10        this.animals = new ArrayList<>();
11    }
12
13    public String getZooName() { return zooName; }
14
15    // Function to add an animal with validation
16    public boolean addAnimal(Animal animal) {
17        if (animal.getName() == null || animal.
18        getName().trim().isEmpty() ||
19        animal.getColour() == null || animal.
20        getColour().trim().isEmpty()) {
21            System.out.println("Error: Name or colour
22            cannot be blank. Animal not saved.");
23            return false;
24        }
25        animals.add(animal);
26        System.out.println(animal.getName() + " added
27        to the zoo.");
28        return true;
29    }
30
31    // Function to Remove animal
32    public void removeAnimal(String name) {
33        boolean found = false;
34        for (int i = 0; i < animals.size(); i++) {
35            if (animals.get(i).getName().
36            equalsIgnoreCase(name)) {
37                animals.remove(i);
38                System.out.println(name + " removed."
39                );
40                found = true;
41                break;
42            }
43        }
44    }
45 }
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```
36         }
37     }
38     if (!found) System.out.println("Animal not
found.");
39 }
40
41 // Function to View all animals
42 public void displayAllAnimals() {
43     if (animals.isEmpty()) {
44         System.out.println("Zoo is empty.");
45         return;
46     }
47     for (Animal a : animals) {
48         System.out.println(a.makeSound());
49     }
50 }
51
52 // Function to Search all animals by name or
colour
53 public void searchAnimal(String query) {
54     boolean found = false;
55     for (Animal a : animals) {
56         if (a.getName().equalsIgnoreCase(query
) || a.getColour().equalsIgnoreCase(query)) {
57             System.out.println("Found: " + a.
makeSound());
58             found = true;
59         }
60     }
61     if (!found) System.out.println("No match
found.");
62 }
63
64 // Function to Perform daily care
65 public void performDailyCare() {
66     for (Animal a : animals) {
67         if (a instanceof Flyable) { ((Flyable) a
).performWingCheck(); }
68         if (a instanceof Swimmable) { ((Swimmable
) a).performWaterQualityCheck(); }
69     }
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70     }
71
72     // Function to Print Zoo Report
73     public void printZooReport() {
74         System.out.println("\n--- Zoo Report: " +
75         zooName + " ---");
76         int eagles = 0, penguins = 0, ducks = 0;
77         HashMap<String, Integer> colours = new
78         HashMap<>();
79
80         for (Animal a : animals) {
81             if (a instanceof Eagle) eagles++;
82             else if (a instanceof Penguin) penguins
83             ++;
84             else if (a instanceof Duck) ducks++;
85
86             String col = a.getColour().toLowerCase
87             ();
88             colours.put(col, colours.getOrDefault(
89             col, 0) + 1);
90         }
91
92         System.out.println("Eagles: " + eagles +
93         " | Penguins: " + penguins + " | Ducks: " + ducks);
94
95         String dominant = "None";
96         int max = 0;
97         for (Map.Entry<String, Integer> entry :
98         colours.entrySet()) {
99             if (entry.getValue() > max) {
100                 max = entry.getValue();
101                 dominant = entry.getKey();
102             }
103         }
104         System.out.println("Dominant Colour: " +
105         dominant + " (" + max + ")");
106     }
107
108     // Function to Modify animal details
109     public void modifyAnimalWeight(String name,
110     double newWeight) {

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102         boolean found = false;
103         for (Animal a : animals) {
104             if (a.getName().equalsIgnoreCase(name
105         )) {
106                 a.setWeight(newWeight);
107                 System.out.println(name + "'s weight
108         has been updated to " + newWeight + "kg.");
109                 found = true;
110                 break;
111             }
112         }
113         if (!found) System.out.println("Animal not
114         found.");
115     }
116     // An additional feature to feed all animals
117     public void feedAnimals() {
118         if (animals.isEmpty()) {
119             System.out.println("No animals to feed!")
120         };
121         return;
122     }
123     System.out.println("\n--- Feeding Time! ---"
124 );
125     for (Animal a : animals) {
126         double weightGained = 0.5;
127         a.setWeight(a.getWeight() + weightGained
128 );
129         System.out.println(a.getName() + " ate
130         their food and gained " + weightGained + "kg! New
131         weight: " + a.getWeight() + "kg.");
132     }
133 }
134
135     // Another additional feature to calculate the
136     total weight of all animals in the zoo
137     public void calculateTotalWeight() {
138         if (animals.isEmpty()) {
139             System.out.println("The zoo is empty.
140         Total weight is 0.0 kg.");
141         return;

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133     }
134     double totalWeight = 0;
135     for (Animal a : animals) {
136         totalWeight += a.getWeight();
137     }
138     System.out.println("\n--- Zoo Logistics ---"
139 );
139     System.out.println("Total combined weight of
    all animals: " + String.format("%.2f", totalWeight
    ) + " kg");
140 }
141
142 // Function to Save data to the 2 files
143 public void saveData() {
144     try (PrintWriter zooOut = new PrintWriter(
145 new FileWriter("ZooDetails.txt"));
146     PrintWriter animalOut = new PrintWriter
147 (new FileWriter("Animal Details.txt"))) {
148
149         zooOut.println(this.zooName);
150         for (Animal a : animals) {
151             if (a.getName().trim().isEmpty() ||
152 a.getColour().trim().isEmpty()) continue;
153
154             if (a instanceof Eagle) {
155                 animalOut.println("Eagle," + a.
156 getName() + "," + a.getAge() + "," + a.getColour
157 () + "," + a.getWeight() + ",2.0");
158             } else if (a instanceof Penguin) {
159                 animalOut.println("Penguin," + a
160 .getName() + "," + a.getAge() + "," + a.getColour
161 () + "," + a.getWeight() + ",15.0");
162             } else if (a instanceof Duck) {
163                 animalOut.println("Duck," + a.
164 getName() + "," + a.getAge() + "," + a.getColour
165 () + "," + a.getWeight() + ",Yellow");
166             }
167         }
168     }
169     System.out.println("Data successfully
    saved to disk.");
170 } catch (IOException e) {

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161         System.out.println("Error saving data."
162     );
163     }
164 }
165 // Function to Load details from disk
166 public void loadData() {
167     try {
168         File animalFile = new File("Animal
169     Details.txt");
170         if (animalFile.exists()) {
171             Scanner fileScanner = new Scanner(
172     animalFile);
173             while (fileScanner.hasNextLine()) {
174                 String[] parts = fileScanner.
175     nextLine().split(",");
176                 if (parts.length >= 6) {
177                     String type = parts[0], name
178     = parts[1], colour = parts[3];
179                     int age = Integer.parseInt(
180     parts[2]);
181                     double weight = Double.
182     parseDouble(parts[4]);
183                     if (type.equals("Eagle"))
184     animals.add(new Eagle(name, age, colour, weight,
185     Double.parseDouble(parts[5])));
186                     else if (type.equals("
187     Penguin")) animals.add(new Penguin(name, age, colour
188     , weight, Double.parseDouble(parts[5])));
189                     else if (type.equals("Duck"
190     )) animals.add(new Duck(name, age, colour, weight,
191     parts[5]));
192                 }
193             }
194             fileScanner.close();
195             System.out.println("Data loaded
196     successfully.");
197         }
198     } catch (Exception e) {
199         System.out.println("No previous data

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187 found or error loading.");  
188     }  
189 }  
190 }
```